The factory

The location of the presentation was quite unusual for the invited guests. Of course, there were many places in the country where such an area could be found: an old factory, or perhaps a mine? The area is a vast, excavated, lunar landscape. After production ceased, looting began at a lower level.
First, the mechanical elements of the buildings, bathroom equipment, lights, electrical wiring, and IT equipment disappeared. Later, the doors, windows, usable roof elements, beams, and then all movable metal structures disappeared. Pipers, hippies, and homeless people invaded the area, and the area was classified as dangerous.

The city jumped at the opportunity when the company showed up and, for a nominal fee, agreed to guard the area and renovate some of the buildings. The guards were recruited locally, and the rules were simple: they were responsible for the vast void between the fence and the building. The fence was high, multi-tiered, and had a few cameras, the guards were ruthless, and the area remained unfriendly and worthless. The debris of the edge of civilization had disappeared as quickly as it had appeared.
No one missed them.

The few selected buildings did not change much from the outside, the buyers did not care about beauty. Behind the old walls, which were considered to be adequate from a static point of view, windowless metal barracks were built under their relative protection, but independently of them. Few people visited here, they arrived by car, from which they only got out inside the gate. They did not even have to interact with people: when they arrived, the appropriate signal appeared on the guards' monitors, the gates opened automatically, and closed behind them.

Many years passed like this, in silence. Then a new, larger hall was built, many people came here by bus, and traffic also increased: trucks came and went. Not with disturbing frequency, but some kind of awakening was felt in the area. In the inner part, areas were fenced off with high, closed walls, from which noises of movement could be heard. Some guards, who were more curious in nature, were dismissed at this time, who dared to approach these walls closer than the designated distance.

Of course, there are always those who want to know everything about everything, but they were easily disarmed: they were shown around the site accompanied by local journalists. The company introduced itself as a company that develops hydraulic and other moving devices, which experiments with new technologies and materials for the unnamed parent company. This explains the small number of employees, the relatively quiet operation - and the understandable mystery. Well, this was not a big number either, it did not excite anyone except for a few eternal and ineradicable fans of conspiracy theories. Those who for some reason were still able to talk to the workers here, cooled the mood further: the buses transported simple workers, engineers, who actually built and tested mechanical devices. And it was really impossible to meet the team of drivers. To this day.

The inventor

The company's chief engineer was a tall, middle-aged man with youthful movements and a spark of obsession in his eyes. He looked as if he had been waiting for this moment for a long time. He quickly turned on the projector behind him and began to speak.

- You have probably seen countless videos circulating on the Internet about various autonomous machines, robot dogs, human-shaped structures, and spider-like machines. The common feature of these is that they use limbs, not wheels, to move around like living creatures. The advantage of rolling is the higher speed, but wheels are not a solution on difficult, rocky terrain, or if you need to climb or jump over an obstacle.

to overcome it.

The common feature of the experiments known today is a certain awkwardness with which these machines move, and that they are basically trying to imitate the movement of some living creature. Engineers choose an animal that they like, take countless recordings of its movements, analyze and break down the steps into parts. Then they build a mechanical model, upload the programs, run them, test, improve them, and modify settings. Despite the investment of a lot of work, the result is worthy of recognition, but… somehow it lacks dynamism.

The problem is that we are too focused on results, too willing to make costly mistakes, too slow to learn. Biological creatures emerged from millions of years of countless failures, and their current efficiency is the result not of design but of constant adaptation and competition. If we want to compete with them, it is not enough to design and copy. We need evolution.

We don't have to make the plans, we have to define the boundary conditions and let the machine figure out how to best use these tools. At least that's what we've been betting on.

In the rooms behind me, life runs on powerful computer simulators, accelerated a million times. You probably have experience with games in which you move around in a virtual physical environment, interacting with various objects, such as car or war simulations. Of course, the emphasis in these is on entertainment, and many simplifications are used for faster and more enjoyable gameplay.

Our systems are more basic, if you will – but far more accurate. There is no need for high-level visualisation, we simulate the environment for the moving, developing, "living" creatures in it with almost complete accuracy. Since these are also computer algorithms, they exist, move and adapt at a pace that is impossible for us to follow. Our job is to model the physical devices available to us as accurately as possible, and here I am talking specifically about motors, pumps, pipes, bearings, frame elements.

The start is simple: we place starting sets in various simulated environments (physical constants, terrain objects, lighting conditions, etc.). We give them tasks to complete based on their individual decisions. The result is automatically evaluated in a complex way, and more successful experimenters can continue trying, while their structure is modified randomly.

Of course, we take into account the limits of what is technically possible, but we do not stop at them: the point is that we also expect the most promising development directions from this environment.

All of this may sound simple, but it is actually terribly complicated, requires huge computational effort, the "tree of possible beings" is too wide, and they have to share finite computational capacity.
We need to provide enough machine time for the “species” to even be able to determine their viability, but random changes simply create too many possibilities, and the system is chaotic: a single experiment cannot be evaluated, the average of thousands of runs can be examined; small changes represent a leap in performance, the result is shaped by the interaction of countless simultaneously existing factors; a minimal change in the evaluation results in a completely different development direction.

This is a very brief summary of our problems, but perhaps it explains why our first few years were a total failure. We were forced to realize that we could only achieve meaningful results with this approach if we had the timescales commensurate with evolution.
Are we a million times faster? But the planet is “running” the program of evolution in a million times more living things, and it has millions of years at its disposal… we don’t. We had to resort to analogies again, in this case, to species breeding.

It may sound strange, but we employed real breeders for our virtual farm. We ourselves approached the task with too much of an engineering, controlling approach, and got stuck on certain parameters. Our breeders brought a new perspective to the system, they were able to look at the whole picture. They demanded that we “see” the creatures in their virtual reality, their bodies, and their movements. The numbers and tables were replaced by concepts such as “beautiful,” “promising,” or even “tucans.” The breeders also worked with target numbers, of course, but they could allocate some of their machine time to their favorites. We also created a kind of competition among them, with more time going to those who had a better “portfolio” – but they had the right to share this time and pass it on to each other.
Breeders were also fired and hired based on their joint decision, we announced tenders, and even developed a game to find talent for them. A strong community spirit developed among them, they exchanged species and further developed each other's stock, as the common goal was obvious: to achieve a "species" that was ripe for real, physical simulation.

Not only us, but also our technologists were made to sweat. Day after day, new demands were made for structural elements, drives, and sensors, which we had to decide how feasible they were. A device with very good parameters but not feasible could have taken machine evolution in a completely wrong direction – just as any possible but rejected development could have cut off the branch leading to ultimate success.

Let me tell you a typical story. The most important motivating factor was, of course, the possibility of developing systems independent of our thinking and habits, which could adapt to conditions that were foreign to us. Our first serious order was to develop a device that could adapt to the Moon, and we owe this entire facility to this plan. The result of the huge investment and long experimentation was a complete failure.

We underestimated the significance of the fact that we can only present the device on a computer. Even though we built it in its physical reality, the programs responsible for its movement are simply inoperable under terrestrial conditions.

I see the astonishment in their eyes: we are programmers, why don't we fix these? The answer is simple: these control systems were not designed by humans, we don't understand them as much as we don't know the nervous system of biological beings, including our own brains, or our thinking enough to make precisely targeted "fixes" to them - and here we are talking about a drastic deviation in a fundamental condition, gravitational attraction. From an earthly perspective, this machine has taken a completely wrong evolutionary path.

It would not have occurred to us to send a pile of metal that had proven perfect in our internal tests but was immobile to the Moon, and then see if it actually worked, and if there were any untested bugs that would require us to experiment with another one. The only option was to build the factory directly on the test site: in this case, on the Moon. The answer is both logical and unfeasible; our project was canceled. Of course, today's technology is again
raises the possibility: if structural elements could be printed with the necessary precision from minerals extracted from lunar dust, and only a few basic components would need to be transported from Earth.

Fortunately, some of our small developments and the capacity of the plant have become marketable so far, and our external projects have generated enough profit to continue the research, albeit with much reduced resources. We have learned our lesson and from now on we will use the simulation set to terrestrial parameters
We worked in an environment so that our results could be presented in physical reality.

You will be the first to see this prototype in its physical form. The task was to move independently, freely, and quickly over difficult terrain, and to transport a forty-kilogram crate. The creation of the structure is, of course, a team effort, the largest part of which belongs to the silent gentleman standing next to me, who created it through hundreds of virtual generations, thousands of pruned branches of development, and years of continuous work in real life… and I can see in his eyes that for him it is not “this”, but “he”, Gron. Don’t ask why. So, he is Gron here.

As we left the building, we stopped in front of a pile of metal about the height of a man. It's hard to describe the effect it had on us. We looked at it in disbelief, several of us smiled mockingly. The structure showed that there must have been an awful lot of work in it, there was a faint humming and gurgling sound coming from it, LEDs were flashing at various points, the whole thing vibrated subtly. Yet, it was a completely meaningless, incomprehensible pile, with the aforementioned crate next to it.

- It's quite large, but in its current state it doesn't even reach six hundred kilograms.
We were able to use lightweight and strong, and therefore expensive, structural elements and components because, unlike our competitors, we had ready-made plans for their construction. The experiments were already underway long before we placed the first order for materials.

In the first part of the presentation, I will demonstrate free movement without load in an extremely simple way. The surface of the tablet I am holding corresponds to our experimental terrain, on which I mark the target with my finger, Gron follows the movement of my hand as long as I continuously touch the screen. If I let go, it stops, if I touch it again further away, it tries to reach the nearest point in the way it has chosen. Of course, it “knows” the terrain, on which, as you can see, there are insurmountable obstacles for it, for example, a pit, a ditch, a few walls. Are you ready? Then let’s go!

Well, we weren't prepared. A motor revved up inside the shapeless mass of metal, and then the whole pile swayed. The sight simply turned my stomach. I had never felt anything like it before, and it seemed that several people shared my fate, and I heard shocked cries. This movement was unlike anything else. It was as if it had tentacles, but no, because in a matter of seconds it turned them inside out, then somehow jerked back, and smote forward again. The whole process was incomprehensible, there was no usual regularity or rhythm to it – and it was terrifyingly fast, as if we had seen a drop of mercury rolling across the terrain.

Then our eyes slowly began to adjust. It became apparent that the unpredictable flapping was actually the result of complete adaptation to the environment. This creature knew every stone and piece of iron in the terrain, and used them to gallop, following the engineer's finger on the screen.
Of course, the control wasn't always perfect - he had obviously intentionally swiped his finger across the ditch, causing the creature to suddenly leap to the side, dodging the ditch at an even faster pace than before, then rejoining the path the finger had taken. Then the engineer lifted it
his finger (the creature crouched), then pointed to a distant point. Gron accelerated and swung with frightening ease across the ditch he had just bypassed, then came to rest at the designated point.

The participants were then given the opportunity to try out the controls. It was a terrifying feeling, but also a childish delight, as the metal ball rolled back and forth across the jagged terrain following the movement of our fingers. It was as if we had changed the gravity and the creature was just rolling up and down following it. Of course, the accessible area was a safe distance from where we were.
We were standing; the experiment started from the edge of it that was falling towards us.

Then came the ability to carry. Gron rolled over to the crate, lifted it with visible effort (and made audible by the noise of the engine), somehow placed it in the center of his body, and then continued.
the demonstration. His movements became noticeably more deliberate, he had to compensate for the inertia of the mass he was carrying. It was shocking to see the… gentleness with which he handled the crate. There was something inhumanly delicate about the way he moved, but the crate remained in place for a moment, then followed the creature's center a little more lazily. It stopped or turned with the same delicacy, and he did not jump the ditch again after that.

- The third part of the presentation may be unusual for you, but in a simulated environment, these too represent a serious selection factor: failures. Machines that work with pre-written, human-designed programs are only able to adapt to a possible structural or control damage to an extremely small extent. Our machines suffer countless "random" failures in their virtual living space, which they have to endure with minimal reduction in their capabilities. All the memories and experiences of each such simulated event are available in the background systems, so the individual, whether virtual or this physical model, quickly adapts to its own changed capabilities. Please put on your earplugs.

The engineer raised a rifle and fired at the creature, which was now pointing towards us from the far side of the field, still carrying the crate. The rough shot had apparently done serious damage to the structure, which momentarily stopped, one might say, fell to its knees, although it continued to hold the crate with careful care. A startled cry broke out from the proud, proud breeder.

It was striking how the LEDs located at different points of the machine suddenly started dancing. “You see, it is now assessing the damage and looking for optimal alternative solutions!” – the engineer said excitedly. And indeed, the machine got up and, although much slower, headed towards us again. The engineer aimed and fired again. The machine shook, its lights began to dance, and it continued to approach, slower and slower, more… pitifully trying to complete the task at all costs.

"Leave the unfortunate man alone..." The soft sigh did not come from the completely collapsed breeder, but from one of the spectators, but it accurately expressed the desire of all of us. It was terrible to watch this suffering, and the soulless precision of the engineer as he slowly executed his victim. Thus, no one was able to comprehend the events of the next moment, let alone react to them.

The engineer raised the weapon again, one of the tentacles of the slowly immobilizing wreck swung... the engineer was hit in the chest by a ten-kilo piece of metal. By the time he fell to the ground, there was no life in him. In the chaos that erupted, we all forgot about the machine for a minute, and when we looked back, it had not moved anymore.

But he carefully placed the box on the target area beforehand.